# Information required for using the snapdrive snap connect command

Snapdrive for Unix

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# Information required for using the snapdrive snap connect command

To connect to a Snapshot copy, determine the type of storage entity, connect a Snapshot copy with the NFS directory tree to Data ONTAP 7.3 configurations, and so on.

The following table gives the information you need to supply when you use the `snapdrive snap connect` command.

> `snap connect` operations require Storage Foundation Enterprise License for `vximport` on Solaris with Veritas.

| Requirement/Option | Argument |
|---|---|
| Decide the type of storage entity that you want to use to attach the Snapshot copy and supply that entity's name with the appropriate argument. This is the value for the *`src_fspec`* argument.<br><br>• If you connect a Snapshot copy of a LUN, SnapDrive for UNIX connects the LUN you specify. You cannot use the `-lun` option on the same command line with the `-vg`, `-dg`, `-fs`, `-lvol`, or `-hostvol` options. You can specify the short name of the LUN in the lun_name or qtree_name/lun_name format.<br>• If you connect a Snapshot copy of a file system that is created directly on a LUN, SnapDrive for UNIX connects the LUN that has the file system.<br>• If you connect a Snapshot copy of a disk group that has a host volume or file specification, the argument translates into a set of disk groups on the storage system. SnapDrive for UNIX connects the entire disk group containing the entity, even if the entity is a host volume or file system.<br>• If you connect a Snapshot copy of an NFS file system, the argument translates to the NFS directory tree. SnapDrive for UNIX creates a FlexClone of the volume, removes directory trees that are not specified in the Snapshot copy, and then connects and mounts the NFS directory tree. If you specify an NFS mount point, you cannot specify non-NFS entities (`-vg`, `-dg`, `-fs`, `-lvol`, or `-hostvol`) on the same command line.<br><br>> SnapDrive for UNIX does not support symbolic links at the mount point level. | |
| LUN (`-lun` *`file_spec`*) | *short name of the LUN.* |
| The *`s_lun_name`* specifies a LUN that exists in the `-snapname` *`long_snap_name`*. The short lun_name is required. You cannot include a storage system or storage system volume name. The *`d_lun_name`* specifies the name at which the LUN is connected. The short *`lun_name`* is required. You cannot include a storage system or storage system volume name. You must specify a *`d_lun_name`* | |
| Disk group (`-dg` *`file_spec`*) or volume group (`-vg` *`file_spec`* ) | *name of the disk or volume group* |
| File system (`-fs` *`file_spec`* ) | *name of the file system* |

| Requirement/Option | Argument |
| --- | --- |
| Host volume (`-hostvol file_spec` ) or logical volume (`-lvol file_spec` ) | *name of the host or logical volume* |
| Connect a Snapshot copy with an NFS directory tree to Data ONTAP 7.3 configurations.<br><br>• If your configuration uses Data ONTAP 7.3 or a later version of Data ONTAP with traditional (not FlexVol) volumes, you must specify this option to connect the Snapshot copy with readonly access (required).<br>• If your configuration uses Data ONTAP 7.3 and later and FlexVol volumes, SnapDrive for UNIX automatically provides read-write access. Specify this option only if you want to restrict access to read-only (optional). | |
| -readonly | *set read only permission* |
| Optional: Supply a name by which the target entity is accessible after the storage entity is connected. SnapDrive for UNIX uses this name to connect the destination entity. This is the *dest_file_spec* argument.<br><br>If you omit this name, the snap connect command uses the value you supplied for *src_fspec*. | |
| Name of target entity | *dest_file_spec* |
| Optional: Specify the names for the destination storage entities. If you included this information as part of the *dest_fspec/src_fspec* pair, you do not need to enter it here.<br><br>You can use the `-destxx` options to specify names for destination storage entities if this information is not part of the *dest_fspec/src_fspec* pair. For example, the `-fs` option names only a destination mount point so you can use the `-destdg` option to specify the destination disk group.<br><br>If you do not specify the name needed to connect an entity in the destination disk group, the `snapdrive snap connect` command takes the name from the source disk group.<br><br>If you do not specify the name needed to connect an entity in the destination disk group, the `snap connect command` takes the name from the source disk group. If it cannot use that name, the operation fails, unless you included `-autorename` at the command prompt. | |
| Destination disk group (`-destdg`) or destination volume group (`-destvg`) | *dgname* |
| Destination logical volume (`-destlv`) or destination host volume (`-desthv`) | *lvname* |
| Specify the name for the Snapshot copy. Use the long form of the name where you enter the storage system name, volume, and Snapshot copy name. | |
| Snapshot copy name (`-snapname`) | *long_snap_name* |

| Requirement/Option | Argument |
| --- | --- |
| `-nopersist` | ~ |
| Optional: Connect the Snapshot copy to a new location without creating an entry in the host file system table.<br><br>• The `-nopersist` option allows you to connect a Snapshot copy to a new location without creating an entry in the host file system table. By default SnapDrive for UNIX creates persistent mounts. This means that:<br>  ◦ When you connect a Snapshot copy on a Solaris host, SnapDrive for UNIX mounts the file system and then places an entry for the LUNs that comprise the file system in the host's file system table.<br>  ◦ You cannot use `-nopersist` to connect a Snapshot copy that contains an NFS directory tree. | |
| `-reserve \| -noreserve` | ~ |
| Optional: Connect the Snapshot copy to a new location with or without creating a space reservation. | |
| Igroup name (`-igroup`) | *`ig_name`* |
| Optional: NetApp recommends that you use the default igroup for your host instead of supplying an igroup name. | |
| `-autoexpand` | ~ |
| To shorten the amount of information you must supply when connecting to a volume group, include the `-autoexpand` option at the command prompt. This option lets you name only a subset of the logical volumes or file systems in the volume group. It then expands the connection to the rest of the logical volumes or file systems in the disk group. In this manner, you do not need to specify each logical volume or file system. SnapDrive for UNIX uses this information to generate the name of the destination entity.<br><br>This option applies to each disk group specified at the command prompt and all host LVM entities within the group. Without the `-autoexpand` option (default), you must specify all affected host volumes and file systems contained in that disk group to connect the entire disk group.<br><br>If the value you enter is a disk group, you do not need to enter all the host volumes or file systems because SnapDrive for UNIX knows what the disk group is connecting to.<br><br>NetApp recommends that, if you include this option, you should also include the `-autorename` option. If the `-autoexpand` option needs to connect the destination copy of an LVM entity, but the name is already in use, the command fails unless the `-autorename` option is at the command prompt. | |
| The command fails if you do not include -autoexpand and you do not specify all the LVM host volumes in all the disk groups that is referred at the command prompt (either by specifying the host volume itself or the file system). | |
| `-autorename` | ~ |

| Requirement/Option | Argument |
| --- | --- |
| When you use the `-autoexpand` option without the `-autorename` option, the `snap connect` command fails if the default name for the destination copy of an LVM entity is in use. If you include the `-autorename` option, SnapDrive for UNIX renames the entity when the default name is in use. This means that with the `-autorename` option at the command prompt, the Snapshot connect operation continues regardless of whether all the necessary names are available.<br><br>This option applies to all host-side entities specified at the command prompt.<br><br>If you include the `-autorename` option at the command prompt, it implies the `-autoexpand` option, even if you do not include that option. | |
| `-devicetype` | ~ |
| Optional: Specify the type of device to be used for SnapDrive for UNIX operations. This can be either “shared” that specifies the scope of LUN, disk group, and file system as host cluster-wide or “dedicated” that specifies the scope of LUN, disk group, and file system as local.<br><br>If you specify the `-devicetype` dedicated option, all the options of snapdrive snap connect command currently supported in SnapDrive 2.1 for UNIX function as they always have.<br><br>If you initiate the `snapdrive snap connect` command with the `-devicetype shared` option from any nonmaster node in the host cluster, the command is shipped to the master node and executed. For this to happen, you must ensure that the `rsh` or `ssh` access-without-password-prompt for the root user should be configured for all nodes in the host cluster. | |
| `-split` | ~ |
| Enables to split the cloned volumes or LUNs during Snapshot connect and Snapshot disconnect operations. | |
| `mntopts` | ~ |
| **Optional:** If you are creating a file system, you can specify the following options:<br><br>• Use `-mntopts` to specify options that you want to pass to the host mount command (for example, to specify host system logging behavior). The options you specify are stored in the host file system table file. Allowed options depend on the host file system type.<br>• The- *`mntopts`* argument is a file system `-type` option that is specified using the mount command `-o` flag. Do not include the `-o` flag in the *`-mntopts`* argument. For example, the sequence -mntopts tmplog passes the string `-o tmplog` to the `mount` command, and inserts the text tmplog on a new command line.<br><br>If you pass any invalid `-mntopts` options for storage and snap operations, SnapDrive for UNIX does not validate those invalid mount options. | |

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